

Zirui Wang

☎ +44 7398 600958 — ✉ ziruiwang.work@gmail.com — [in](#) [LinkedIn](#)

WORK EXPERIENCE

System AI Research Engineer

Huawei Technologies R&D (UK) Ltd

Edinburgh, UK

Oct 2025 – Present

- Designed an agentic AI workload scheduling policy for vLLM-Ascend serving concurrent coding agents (Best-of- N , RL rollouts): lifts prefix-cache hit rate from 56% to 72%, halves queue wait, and completes agent cohorts 17% sooner at equal task-resolve rate, with +9 pts accuracy at the highest concurrency tested. **First-author paper in preparation for ACM SoCC 2026.**
- Engineered a cross-layer observability stack for agentic LLM serving (OpenTelemetry, Prometheus, custom span converters and Gantt exporters), unifying agent-, serving-, and NPU-level telemetry on a single timeline to reveal where accelerator time and KV-cache capacity go during multi-agent runs.

Technical Insight Specialist

Huawei Technologies R&D (UK) Ltd

Edinburgh, UK

Nov 2024 – Oct 2025

- Led comparative evaluations of LLM inference across Cerebras wafer-scale, NVIDIA GPU, and Huawei Ascend architectures, weighing memory and interconnect trade-offs and scale-up versus scale-out designs to inform Huawei's inference infrastructure strategy.
- Proposed model routing and tiered KV-cache storage as research directions ahead of their industry mainstreaming; helped pivot the team's research focus toward agentic AI serving.
- Initiated and designed an agentic AI hackathon on openEuler OS in partnership with the University of Edinburgh, authoring all challenge problems and attracting 100+ student participants.

PROJECT EXPERIENCE

Personalized Dialogue Generation with Implicit Persona Modeling

NLP, In-Context Learning, Dialogue Systems

Mar 2024 – Sep 2024

UCL MSc Dissertation

- Engineered an in-context learning pipeline that infers user persona from dialogue history alone rather than from predefined profiles; outperformed most existing implicit- and explicit-persona systems on fluency and diversity across GPT-3.5/4o and 6 open LLMs.
- Proposed Average Drift Score, a turn-based metric quantifying topic drift; showed drift tendency is consistent within speakers and contagious across conversation partners.

Scientific Data Compression with Variational Autoencoder

PyTorch, VAE, Entropy Coding

Oct 2022 – Jun 2023

UoL BEng Dissertation

- Re-implemented Ballé et al.'s scale-hyperprior VAE for lossy compression of Overthrust 3D seismic waveform data (91K+ datapoints), reaching low-bitrate operating points unreachable by the ZFP baseline at matched distortion.

EDUCATION

University College London

MSc Integrated Machine Learning Systems – **Distinction**

London, UK

Sep 2023 – Sep 2024

University of Liverpool

BEng Computer Science & Electronic Engineering – **First Class Honours**

Liverpool, UK

Sep 2020 – Jul 2023

SKILLS

ML & AI: LLM Inference Optimization (batching, prefix caching, KV-cache mgmt), Agentic AI Serving, PyTorch
Systems & Infra: vLLM / vLLM-Ascend, CrewAI, OpenTelemetry, Prometheus, Docker, Kubernetes
Hardware: Ascend NPU/CANN, NVIDIA GPU & Cerebras wafer-scale, memory/interconnects (HBM, HCCS, CPO)
Programming: Python, C++, Java, SQL, Linux/Unix Shell, Git
Languages: Chinese (Native), English (Fluent), Japanese (Conversational)
Interests: Hardware architecture deep dives, AGI & post-AGI futures, equity & options trading, quantitative analysis